CSE470: Software Engineering

Software Architecture

What is Software Architecture

Architectural design is concerned with understanding how a software system should be organized and designing the overall structure of that system.

Software Architecture is how the defining components of a software system are organized and assembled. How they communicate each other. And how the constraints of the whole system is ruled by.

The architectural model serves as a input to the development phase



Advantages of explicit architecture

Stakeholder communication

- Architecture may be used as a focus of discussion by system stakeholders.

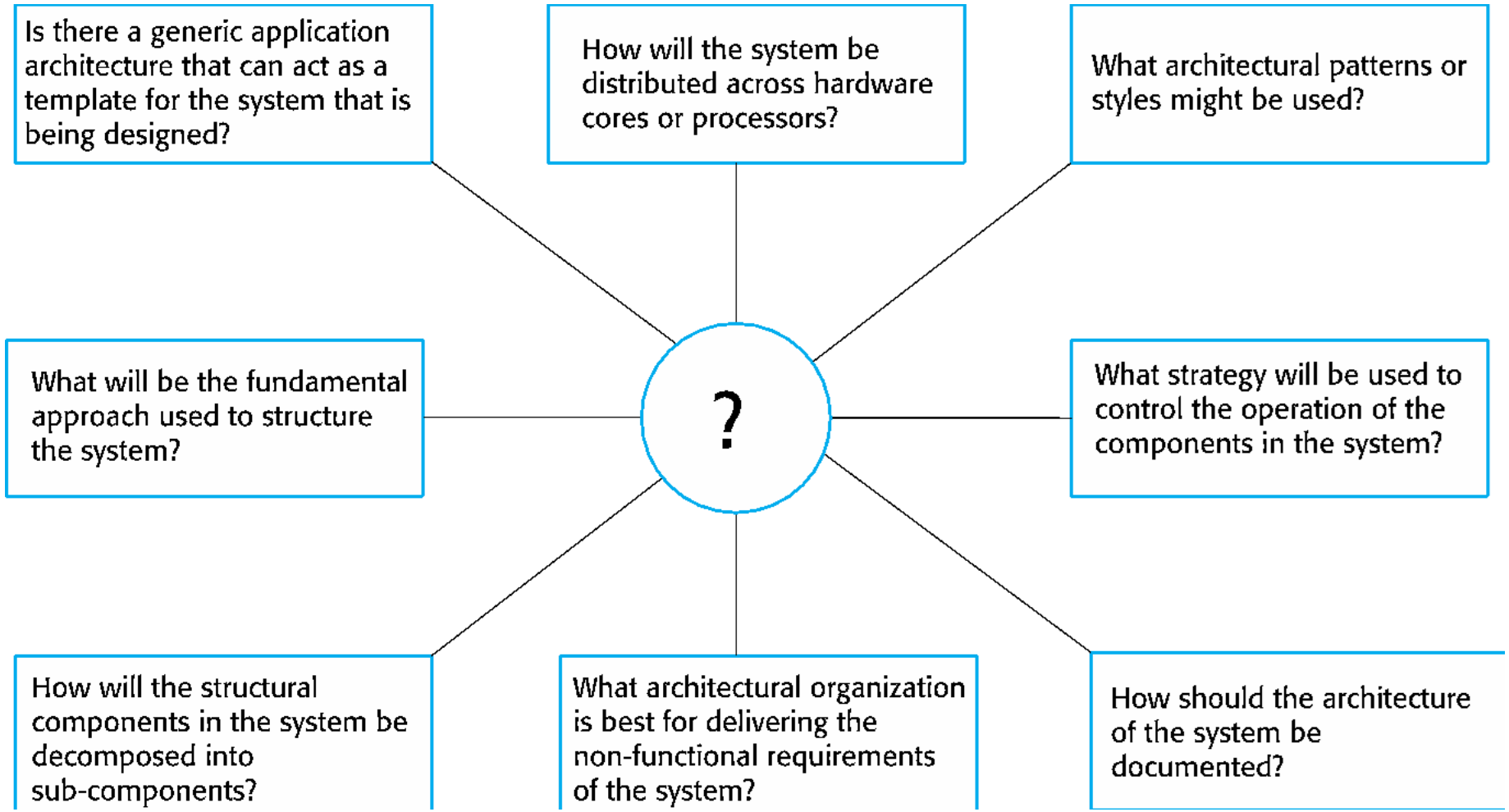
System analysis

- Means that analysis of whether the system can meet its non-functional requirements is possible.

Large-scale reuse

- The architecture may be reusable across a range of systems
- Product-line architectures may be developed. (Softwares with common features like e-commerce)

Architecture Design Decisions



Architectural View

- There are four views from which the architecture of a software can be observed
- Each architectural diagram only shows one view or perspective of the system.

